MILITARY SPECIFICATION

RECORDER-PROCESSOR-VIEWER, RADAR MAPPING

RO-495( )/U

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Recorder-Processor-Viewer, Radar Mapping RO-495( )/U used with Radar Surveillance Sets AN/APS-94E and AN/APS-94F, and Receiving Set, Radar Data AN/TKQ-2B.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items
MIL-F-14072	- Finishes for Ground Electronic Equipment
MIL-P-15024/6	- Plate, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-E-55585	- Electronics Equipment and Parts, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5841

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurements of
MIL-STD-726	- Packaging Requirements Code
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems Equipment and Facilities

DRAWINGS

SM-B-878155	- Recorder-Processor-Viewer, Radar Mapping RO-495/U
SM-A-878404	- Burn-in Test Procedure RO-495/APS-94E Recorder Processor Viewer
SM-A-878408	- Acceptance Test Procedure for Radar Surveillance Set AN/APS-94E

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The Recorder-Processor-Viewer, Radar Mapping RO-495( )/U shall be constructed in accordance with the requirements of this specification and of SM-E-878155 (see 4.13).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply (see 4.4).

3.4 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

3.5 Finish. All surfaces of the Recorder-Processor-Viewer, Radar Mapping RO-495( )/U requiring a protective coating shall be finished in accordance with MIL-F-14072 and the equipment drawings specified herein (see 4.4).

3.6 Nameplate and marking.

3.6.1 General. Marking of Recorder-Processor-Viewer, Radar Mapping RO-495( )/U shall be in accordance with MIL-M-13231. Front panel marking shall be group I (see 4.4).

3.6.2 Visibility. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipments.

3.6.3 Nameplates. Each nomenclatured item shall have a nameplate conforming to MIL-P-15024/6, Type H or B (see 4.4).

3.6.4 Serial numbers. Each unit of Recorder-Processor-Viewer, Radar Mapping RO-495( )/U shall be serial-numbered (see 4.4).

3.7 Systems safety engineering.

3.7.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454 and paragraph 5.13 of MIL-STD-1472. Compliance with these requirements will be verified through a visual inspection (see 4.13).

3.7.2 Corner and edge rounding. All exposed corners and edges shall be rounded to minimum of .75mm (.03 in) radius. Sharp edges and corners that present a personnel safety hazard or potential damage to equipment during use shall be suitably protected or rounded to a minimum radius of 13mm (0.5 in).

3.7.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.8 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.9 Weight. The RO-495( )/U weight shall not exceed 110 lbs (49.9 Kg) (see 4.7).

3.10 Burn-in. The RO-495( )/U shall meet the requirements herein, without subsequent processing after subjection to the burn-in of 4.6.

3.11 Performance requirements. The RO-495( )/U shall meet the specified performance requirements of SM-A-878408 (see 4.9).

3.11.1 Compatibility. The RO-495( )/U shall be compatible electrically and mechanically with Radar Surveillance Set AN/APS-94E and AN/APS-94F (see 4.9.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

3.12 Service conditions. The RO-495( )/U shall meet the following service conditions (see 4.10).

3.12.1 Temperature and altitude. The RO-495( )/U shall operate over a temperature range of -40°F (-40°C) to +131°F (+55°C) with intermittant operation to +160°F (+71°C), and under combined temperature-altitude conditions in accordance with category 3, table 504.1-II of MIL-STD-810 except when processing film the temperature limits shall be 41°F (+5°C) to 106°F (+38°C) preceded by step 1a of Category 2, at altitudes from sea level to 15,000 feet (4,592m). In addition, the equipment shall withstand temperature storage over a range of -80°F (-62°C) to +185°F (+85°C), and air transport at altitude to 40,000 feet (12,192m).

3.12.2 Humidity. The RO-495( )/U shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme humidity levels as encountered in tropical areas (see 4.10.2).

3.12.3 Salt fog. After the salt fog test of 4.10.3, the RO-495( )/U shall operate without degradation in specified performance and show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface exfoliation (see 6.4). There shall also be no clogging or binding of moving mechanical parts.

3.12.4 Fungus. After the fungus test of 4.10.4, the RO-495( )/U shall be visually examined. The RO-495( )/U shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the RO-495( )/U shall operate without degradation in specified performance after exposure to the test conditions of 4.10.4. Materials used in the manufacture, fabrication, construction and treatment of the RO-495( )/U shall comply with Requirement 4 of MIL-STD-454.

3.12.5 Vibration. The RO-495( )/U shall operate without degradation in specified performance and shall not suffer any mechanical damage after being subjected to the vibration test specified in 4.10.5.

3.12.6 Shock. The RO-495( )/U shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the shock test specified in 4.10.6.

3.12.7 Explosive atmosphere. The RO-495( )/U shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere (see 4.10.7).

3.12.8 Bench handling. The RO-495( )/U shall be operable without degradation in specified performance after being subjected to the test of 4.10.8.

3.12.9 Crash safety. The RO-495( )/U shall be operable without degradation in specified performance and shall suffer no mechanical damage after being subjected to the crash safety test specified in 4.10.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

3.12.10 Dust. The RO-495( )/U shall operate without degradation in specified performance after being subjected to the dust test of 4.10.10.

3.13 Electromagnetic interference. The RO-495( )/U shall meet the requirements of MIL-STD-461A, Notice 4 (see 4.11).

3.14 Reliability. The RO-495( )/U shall have a specified (O_o) mean-time-between-failure (MTBF) of 120 hours the lower test MTBF (O_1) shall be 60 hours. The reliability requirements shall be demonstrated under the conditions specified in 4.12.

3.15 Workmanship. The RO-495( )/U shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.13):

General requirements for plastic material and plastic parts
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Soldering and brazing
Cleaning
General requirements for securing of parts
Cleaning prior to welding
Process for welding
Controls
Self-tapping screws
Cleaning after welding
Printed wiring assemblies

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

Quality conformance inspection of equipment before packaging (see 4.5).
Packaging inspection (see 4.14).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

4.3.1 First article units. The contractor shall furnish five first article units of the RO-495( )/U for group A, B, C and D testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of tests: Unit 1	Unit 2	Unit 3	Units 4-5
1. Inspections covered by subsidiary documents 1/	3.3 3.5 3.6 3.15	4.4	Inspection to be performed on all units			
2. Burn-in 1/	3.10	4.6	Tests to be performed on all units			
3. Group A inspection 1/	See table II		Tests to be performed on all units			
4. Group B inspection 1/	See table III		Tests to be performed on all units			
5. Group C inspection						
Temperature-altitude	3.12.1	4.10.1	1			
Humidity	3.12.2	4.10.2	2			
Salt fog	3.12.3	4.10.3		3		
Vibration	3.12.5	4.10.5			1	
Shock	3.12.6	4.10.6			2	
Fungus	3.12.4	4.10.4			3	
Explosive atmosphere	3.12.7	4.10.7		2		
Bench handling	3.12.8	4.10.8			4	
Crash Safety	3.12.9	4.10.9			5	
Dust	3.12.10	4.10.10	3			
6. EMI	3.13	4.11		1		
7. Group D inspection Reliability	3.14	4.12				1

1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.5
Nameplate and marking	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4, and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL major	AQL minor
Visual and mechanical	3.15	4.13	1%	4%
		4.14	1%	4%
System safety	3.7	4.13	1%	4%
Performance	3.11	4.9	1%	1/

1/ All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sample plans, for the AQL's listed in table III, shall be inspection level S-4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Interchangeability	3.8	4.8	6.5%
Weight	3.9	4.7	6.5%
Compatibility	3.11.1	4.9.1	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on inspection lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. Three RO-495( )/U shall be selected at random, one sample for each subgroup I, II and III in table IV, from every 25 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot. The salt fog, electromagnetic interference and fungus tests shall be performed only on the first sample units selected for the first group C inspection.

4.5.3.2 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Temperature-altitude	3.12.1	4.10.1
Humidity	3.12.2	4.10.2
Dust	3.12.10	4.10.10
Subgroup 2		
EMI	3.13	4.11
Explosive atmosphere	3.12.7	4.10.7
Salt fog 1/	3.12.3	4.10.3
Subgroup 3		
Vibration	3.12.5	4.10.5
Shock	3.12.6	4.10.6
Fungus 1/	3.12.4	4.10.4
Bench handling	3.12.8	4.10.8
Crash safety	3.12.9	4.10.9

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.4 Group D inspection. This inspection shall consist of the test specified in table V and shall be performed on units from lots which have been subjected to and met group A and group B inspections.

4.5.4.1 Sampling for group D inspection. Two samples of RO-495( )/U shall be selected at random for group D inspection from every 50 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot.

TABLE V. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.14	4.12

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

4.5.4.2 Group D failures. In case of group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.4.3 Reinspection of group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

4.6 Burn-in. Each RO-495( )/U shall be subjected to the burn-in requirements of SM-A-878404 prior to being subjected to group A testing (see 3.10).

4.7 Weight. The RO-495( )/U shall be weighed on a calibrated scale to determine compliance with the weight requirement of 3.9.

4.8 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the units. Mechanical dimensions and electrical performance required by the RO-495( )/U drawings shall be measured to determine compliance with Requirement 7 of MIL-STD-454 (see 3.8).

4.9 Performance testing. The RO-495( )/U shall be tested for compliance with specified performance requirements of SM-A-878408, Acceptance Test Procedure for Radar Surveillance Set AN/APS-94E (see 3.11).

4.9.1 Compatibility. The compatibility of the RO-495( )/U shall be demonstrated by performing an operational test as part of Radar Surveillance Set AN/APS-94E or AN/APS-94F (see 3.11.1).

4.10 Service conditions. The service condition tests shall be performed as detailed below. Each RO-495( )/U subjected to these tests shall have passed group A tests and shall be resubmitted to and pass these group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass an operational test of 4.9.

4.10.1 Temperature-altitude. The RO-495( )/U shall be subjected to the test of Method 504.1, Procedure I of MIL-STD-810 in accordance with category 3 of Table 504.1 - II, except when processing film, shall be in accordance with category 2 (see 3.12.1).

4.10.2 Humidity. The RO-495( )/U shall be subject to the test MIL-STD-810, Method 507.1, Procedure II. Prior to measurements, excess surface water may be removed from the equipment by wiping external surfaces only. In step 6, operation of 4.9 measurements shall be performed during the last 5 hours of the last cycle (see 3.12.2).

4.10.3 Salt fog. The RO-495( )/U shall be tested in accordance with Method 509.1, Procedure I of MIL-STD-810. Inability of the RO-495( )/U to meet the requirements of 3.12.3 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

4.10.4 Fungus. The RO-495( )/U shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the RO-495( )/U is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the RO-495( )/U. Inability of the RO-495( )/U to meet the requirements of 3.12.4 shall constitute failure of this test.

4.10.5 Vibration. The RO-495( )/U mounted with vibration isolators in place, shall be subjected to the test method 514.2 of MIL-STD-810, procedure I, category b.I, curve C, of figure 514.2-2 (Part 1) and also Curve B of figure 514.2-2 with isolators removed. Sinusoidal vibration levels shall be subject to acceleration weight derating in accordance with paragraph 4.5.1 of Method 514.2. The maximum frequency shall be 500 Hz, and the sweep time shall be 30 minutes. The equipment shall be operating during all vibration testing (see 3.12.5).

4.10.6 Shock. The RO-495( )/U shall be subjected to the test of Method 516.2, Procedure I, of MIL-STD-810, using the half-sine pulse of Figure 516.2-2. Equipment need not operate during shock test. Inability of RO-495( )/U to meet the requirements of 3.12.6 shall constitute failure of this test.

4.10.7 Explosive atmosphere. The RO-495( )/U shall be subjected to the test of MIL-STD-810, Method 511.1, Procedure I (see 3.12.7).

4.10.8 Bench handling. The RO-495( )/U shall be subjected to the tests of Method 516.2, Procedure V, of MIL-STD-810 (see 3.12.8).

4.10.9 Crash safety. The RO-495( )/U shall be subjected to the tests of method 516.2, Procedure III, of MIL-STD-810, using the half-sine pulse of Figure 516.2-2. The equipment shall not be operated during the test (see 3.12.9).

4.10.10 Dust. The RO-495( )/U shall be subjected to the tests of Method 510.1, Procedure I, of MIL-STD-810, except that the air velocity in steps 1, 2, and 3 shall be 200 $\pm$ 100 feet per minute. Step 3 shall proceed immediately after stabilization in Step 2 (see 3.12.10).

4.11 Electromagnetic interference. The RO-495( )/U shall be tested for compliance with the requirements of MIL-STD-461A, Notice 4, using the approved test methods of MIL-STD-462, Notice 3. The applicable requirements of that standard are:

CE01	CS01	RE02	RS02
CE02	CS02	RE02.1	(a) RS03
CE04	CS06		

The RO-495( )/U shall not malfunction, exhibit degraded performance or undesirable response when immersed in electric fields of the following intensity using the modulation characteristic specified:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

Frequency range	Field intensity	Modulation characteristics
10KHz - 2MHz	1 volt/meter	AM, 50%, 1000Hz tone
2MHz - 30MHz	5 volt/meter	AM, 50%, 1000 Hz tone
30MHz - 76MHz	10 volt/meter	FM, + 8kHz deviation, 1000Hz tone
76MHz - 400MHz	10 volt/meter	AM, 50%, 1000Hz tone
400MHz - 2GHz	10 volt/meter	Pulse, 0.1 usec pulse width, 800 pps
2GHz - 12.4GHz	5 volt/meter	Pulse, 0.1 usec pulse width, 800 pps

4.12 Reliability tests.

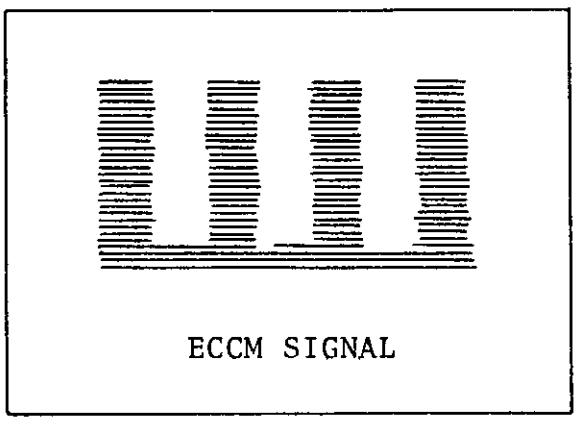
4.12.1 Reliability testing. To determine compliance with 3.14, the RO-495( )/U shall be tested in accordance with test plan XIIC of MIL-STD-781 and the government approved test plan and procedure as required in the contract.

4.12.2 Reliability test conditions. The RO-495( )/U shall be operated under the conditions specified for burn-in (4.6).

4.12.3 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from the performance as required by this specification.

4.12.4 Parameters to be measured. The following performance parameters shall be performed at least once per day on each equipment during the reliability test:

a. Observe the proper display on the monitor scopes (reference the following figure).



b. The 5 MHz test signal should be present on both the FT and MT channels during the period that the test video control was switching to ON.

c. The width of the zone having to test signals should be between 8/16 and 9/16 of an inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

4.13 Visual and mechanical. The RO-495( )/U shall be examined for the defects listed in MIL-STD-252. In addition, RO-495( )/U shall be visually inspected to determine conformance to the requirements of 3.7.

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Material inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior marking shall be in accordance with group A and group B quality conformance inspection requirements of MIL-P-116: lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and marking for shipment and storage shall consist of the examination specified in table VI. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE VI. Packing inspection provision.

Characteristic	Inspection
Intermediate container not as specified	Visual
Improper closure of intermediate container	Visual
Shipping containers not in accordance with specification	Visual
Excessive cube	Visual
Improper blocking and bracing	Visual
Closure not in accordance with specification	Visual
Weight and size exceed container limitations	Weight & Measure
Strapping not in accordance with specification, incorrectly applied, omitted	Visual
Marking omitted, incorrect, or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 4Ø-1-1-ØØ-00-NR-X-ED-Ø-00-A.

5.1.2 Level B. 3Ø-1-1-ØØ-00-NR-X-ED-Ø-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The RO-495( )/U part of AN/APS-94E and AN/APS-94F is intended for use in the Mohawk aircraft and the AN/TKQ-2B ground sensor terminal. Two separate maps are produced simultaneously on the same strip of film by the Recorder-Processor-Viewer, Radar Mapping RO-495( )/U (recorder). One map provides a record of fixed target (FT) information (radar returns from all targets). The other map provides a record of moving target (MT) information (radar returns from moving targets).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see section 5).

c. When first article inspection rough handling tests are required (see section 4).

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

h. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design, for example, RO-495A/U. The contractor shall apply for nomenclature in accordance with applicable clause in the contract (see 1.1).

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

Custodian:
Army - ER

Preparing Activity:
Army - ER
Project 5841-A027

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 6, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY DEPARTMENT OF THE ARMY

Commander
US Army Combat Surveillance and
Target Acquisition Laboratory
ATTN: DELCS-PE
Fort Monmouth, New Jersey 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-49444(ER)	Recorder-Processor-Viewer, Radar Mapping RO-495( )/U

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2020-10-22T20:00Z
Check the source to verify that this is the current version before use.